Leo Zhou

346-859-1973 zhouleo2007@gmail.com leozhou8 leozh0u Portfolio

Education

Rice University | B.S. in Computer Science | **GPA**: 3.92/4.00 **Expected May 2028**

- **Relevant Coursework**: Data Structures & Algorithms, Linear Algebra, Computer Systems, Program Design, Compilers, Probability & Statistics, Discrete Math, Real Analysis, Computer Engineering, Digital Logic Design
- **Activities and Societies**: Competitive Programming Club, Chess Club, Rice Fencing Club, Rice Wind Energy

Work Experience

Software Engineer | **KEWW** | Westchase Hospitality Group **September 2026 – Present**

- Built Cashew in Next.js/TypeScript to replace manual Caterease entry with reviewed Gemini intake, source quotes, deterministic pricing and six linked documents; estimated preparation time: **30 minutes to under 1 minute**
- Building a client portal backed by Postgres row-level security to replace HubSpot forms and manual re-entry; clients can access prices, proposals, menus and event details to reduce routine calls and emails
- Indexing 200,000 email threads and 15 years of records into Postgres with pgvector, PII redacted by Presidio
- Designing SEO and marketing agents to draft reviewed web pages and campaigns from approved sources

Software Developer | **RiceApps × Rice Football** **September 2026 – Present**

- Building Rice Football's data backend to unify PFF, CFBD and Hudl feeds in a shared, source-tagged database
- Developing Python/R packages to query shared data and reuse analyses across Rice Football coaching reports
- Designing cloud execution of analyst scripts to deliver fresh PDF reports on demand to coaching staff

Undergraduate Research Assistant | **Rice University** | Prof. Jiarong Xing **August 2026 – Present**

- Investigating how **colluding AI agents** can bypass access policies for a systems-security research proposal
- Fixed OAuth refresh and Keychain permissions in TokenGate, a Swift macOS app for Claude usage tracking; **eliminated daily re-logins** and added automatic SSH sync to track usage from remote sessions

Software Engineer Intern | **CanSemi** | Test Engineering Team **June 2026 – August 2026**

- Built a Python/PostgreSQL test-data pipeline used by six engineers to replace manual log parsing; normalized 500+ binary ATE logs and processed 40 wafer lots (∼12,000 records) in **under 1 minute**
- Exposed test results through 8 FastAPI endpoints, cutting engineers' report turnaround by ∼**75%**
- Built a React/Canvas wafer map with hover and drill-down to inspect test results, replacing Excel heatmaps

Software Engineer Intern | **Inkstone Technologies** **April 2026 – June 2026**

- Built a SwiftUI app with Apple Vision OCR and a per-user character model to annotate unfamiliar Chinese text
- Parsed and frequency-ranked a 113,000-entry Chinese dictionary in Python into an indexed SQLite database

Projects

BlunderNet | blundernet.com | *Go, C++, React, PostgreSQL, Redis, AWS, Terraform, Docker* **Ongoing**

- Built a chess training site for ∼200 users, with 3.25M puzzles filtered by rating, theme and game phase
- Cut uniform puzzle selection from **1.4 s to 0.9 ms** using indexed shuffle keys and size-weighted cells
- Built a C++/pybind11 MCTS core with batched inference, searching **1.8× faster** than Python
- Built Terraform-defined stateless Go APIs, Redis state and SQS-fed ONNX workers on AWS; load-tested 270/270 replies at 1.5 games/s (1.25 s p95), with queue-based scaling from one to four workers

Vestigo | vestigo.earth | *Python, PyTorch, CLIP/SigLIP, tool calling, calibration* **Ongoing**

- Built a photo geolocation agent that validates tool citations and limits location precision to its evidence
- Trained a PyTorch SigLIP classifier on 64,865 images split by location; reduced median error from 198 km with CLIP ViT-L/14 to **142 km**, with 51.6% cell accuracy and **3.7% calibration error** after temperature scaling

From Scratch | fromscratch | *TypeScript, React, Tailwind, Vite, Gemini API* **September 2026**

Leadership & Activities

Rice Fencing Club | Co-president, Coach & Website Manager | Three SWIFA wins **August 2025 – Present**

- Built and manage ricefence.com with scroll-driven animations for recruitment, photos and club info

Junior National Fencing Team | Captain, Coach & Referee | Junior Worlds + Commonwealth **2022 – 2025**

- Won 30+ titles across New Zealand and Australia, coached 30+ young fencers and refereed 50+ bouts

Physics Pathfinders | Charity President | Physics workshops for 600+ students across 22 schools **2023 – 2025**

Technical Skills

Languages: Python, C++, Go, TypeScript/JS, SQL, C, HTML/CSS | **Spoken**: English, Mandarin, Spanish
Frameworks & Data: React, Next.js, FastAPI, SwiftUI, PyTorch, PostgreSQL, Redis, SQLite, pgvector
Systems & Tools: AWS (ECS, SQS, RDS), Terraform, Docker, Linux, Git, GitHub Actions, WebSockets, ESP32